

A continuous function with zero weak gradient is constant

MathlibAnnex.WeakDivergenceZero.exists_eqOn

theorem

Removes the exceptional set using continuity and positivity of open sets.

Statement

Let E be a finite-dimensional real normed vector space with its Borel structure and an additive Haar measure μ . Let $U \subseteq E$ be open and preconnected. Let $u : E \rightarrow \mathbb{R}$ be locally integrable and continuous on U , and suppose

$$\int_U u(z) \operatorname{div} W(z) d\mu(z) = 0$$

for every C^1 vector field $W : E \rightarrow E$ for which a compact set K_W satisfies

$$\overline{\{z \in E : W(z) \neq 0\}} \subseteq K_W \subseteq U.$$

Here $\operatorname{div} W(z) := \operatorname{tr}(DW(z))$. Then there is $c \in \mathbb{R}$ such that

$$u(x) = c \quad \text{for every } x \in U.$$

Assumptions

Preconnectedness means that U has no partition into two disjoint nonempty relatively open subsets; the empty set is allowed. In addition to openness, this condition, local integrability on U , and the weak equation, continuity is required only on U . The ambient space may have dimension zero. No global regularity, boundedness, or normalization of Haar measure is required.

Conclusion

The same constant obtained by the almost-everywhere theorem works pointwise on U . There is no assertion about values outside U .

Pointwise equality and almost-everywhere equality are different conclusions. The upgrade here uses both continuity and the positive measure of every nonempty open subset; the weak equation alone does not justify changing values on an exceptional set into pointwise information.

Proof route

First obtain a common constant almost everywhere. If a point had a different value, continuity would create a nonempty open neighborhood of different values, contradicting almost-everywhere equality.

▼ Proof steps

1. **Fix the almost-everywhere constant.** Apply the global almost-everywhere constancy theorem to openness, preconnectedness, local integrability, and the weak equation. It gives a real c with

$$u = c \quad \mu|_U\text{-almost everywhere.}$$

This step does not use continuity.

2. **Exclude any exceptional point.** The open-set continuity lemma applies to u and the constant function with this same c : both are continuous on the open set U , they agree almost everywhere there, and every nonempty open set has positive measure. Its contradiction argument is as follows. Suppose $x \in U$ and $u(x) \neq c$. Set $\delta = |u(x) - c|/2 > 0$. Continuity on the open set U provides an open neighborhood V of x , contained in U , such that $|u(y) - u(x)| < \delta$ for all $y \in V$. Then

$$|u(y) - c| \geq |u(x) - c| - |u(y) - u(x)| > \delta \quad (y \in V).$$

The almost-everywhere equality restricted to V says $u = c$ there almost everywhere. But V is nonempty and open, so $\mu(V) > 0$ and it contains a point satisfying that equality, a contradiction. Thus no such x exists and $u = c$ on U . If U is empty, the pointwise conclusion is vacuous.

Main citations

- Exact declaration and its source — `MathlibAnnex.WeakDivergenceZero.exists_eqOn`
- The same weak hypotheses give an a.e. constant — `MathlibAnnex.WeakDivergenceZero.exists_aeConstantOn`
- Equality everywhere from a.e. equality on an open set — `MeasureTheory.Measure.eqOn_open_of_ae_eq`

Lean source signature (exact)

```
theorem WeakDivergenceZero.exists_eqOn
  {U : Set E} {u : E → ℝ} (hweak : WeakDivergenceZero μ U u)
  (hU : IsOpen U) (hUc : IsPreconnected U) (hu : LocallyIntegrableOn u U μ)
  (hcont : ContinuousOn u U) : ∃ c : ℝ, Set.EqOn u (fun _ => c) U
```

In the source	Mathematical meaning
<pre>[NormedAddCommGroup E] [NormedSpace ℝ E] [FiniteDimensional ℝ E] [MeasurableSpace E] [BorelSpace E] {μ : Measure E} [μ.IsAddHaarMeasure]</pre>	The finite-dimensional real normed space E, with Borel measurable structure and additive Haar measure μ. Dimension zero is allowed.
<pre>(hweak : WeakDivergenceZero μ U u)</pre>	The scalar weak equation $\int_U u(z) \operatorname{div} W(z) d\mu(z) = 0$ for every C^1 vector field with a compact carrier contained in U; $\operatorname{div} W = \operatorname{tr}(DW)$.

In the source	Mathematical meaning
<code>(hU : IsOpen U) (hu : LocallyIntegrableOn u U μ)</code>	The set U is open and u is locally integrable on U . Whole-space integrability or Lipschitz continuity of u is not assumed.
<code>(hUc : IsPreconnected U)</code>	The domain is preconnected and may be empty.

In the source	Mathematical meaning
<code>(hcont : ContinuousOn u U)</code>	The extra assumption is ordinary continuity of u on U , not a.e. continuity or global continuity.
<code>∃ c : ℝ, Set.EqOn u (fun _ => c) U</code>	There exists one real c with $u(x) = c$ for every $x \in U$. <code>fun _ => c</code> is the constant function and <code>EqOn</code> quantifies pointwise on U .

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 7–12:

```
noncomputable section
open Set MeasureTheory TopologicalSpace
namespace MathlibAnnex
variable {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
  [FiniteDimensional ℝ E] [MeasurableSpace E] [BorelSpace E]
variable {μ : Measure E} [μ.IsAddHaarMeasure]
```

▼ Exact content identity

Declaration: `MathlibAnnex.WeakDivergenceZero.exists_eqOn`

Accepted content SHA-256: `c3b9910b26fee458232ba0247b83cbe8decac623616d870480cff8481d8c2e6`

Accepted source guide SHA-256: `f8158618bd8f1f8986c7790011350a07218014fb772889a6977df8bc59261760`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`